TECHNICAL DILIGENCE REVIEW - RECORD 013

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-013

1. Scope, targets and the basis of the capacity assessment

The present report was prepared by Wrenfield Technical Advisory LLP for Ardent Peak and covers the capacity position, load-test evidence and latency behaviour of the platform at fleet scale at Helioscope Robotics, Inc. as at 2026-02-18. As instructed, Wrenfield Technical Advisory LLP confined this report to the capacity position, load-test evidence and latency behaviour of the platform at fleet scale; three adjacent questions raised during the work are listed as follow-ups rather than answered here. The principal limitation on TDR-2026-02 is time: 3 weeks was enough to test the control path and not enough to test the reporting path. The reviewer notes that the present report was prepared while three remediation programmes were in flight, so several positions will have moved by completion. The numbering in TDR-2026-02 follows the order of the work rather than the order of importance.

The sample behind the present report is 9 services and 6 sites; the selection was the reviewer's, not the company's. 9 services were examined in detail for this report, chosen for their position in the dependency map rather than at random. Scope of the present report is the capacity position, load-test evidence and latency behaviour of the platform at fleet scale, and the cut-off for evidence is 2026-02-18. The subject of TDR-2026-02 is the capacity position, load-test evidence and latency behaviour of the platform at fleet scale, examined for Ardent Peak by Wrenfield Technical Advisory LLP over 3 weeks ending 2026-02-18. TDR-2026-02 does not restate the company's own reporting where the reviewer was able to verify the underlying data instead.

Evidence gathering for this report was iterative: an initial document request, 16 interviews, then a second request that closed most of the gaps. Access for this report was sufficient for the conclusions drawn; where it was not, the limitation is stated at the point it bites. Wrenfield Technical Advisory LLP recommends that the 6 indicators be reported with their measurement basis attached, since two of them changed basis during the reviewed period. L. Casaubon would add one qualitative item to the pack proposed in the present report: the count of commitments closed without verifying evidence. This report was prepared on the footing that a reader may be an engineer or may not, and defines terms where the distinction matters.

In TDR-2026-02 the reviewer records the basis of each conclusion inline, so a reader can disagree with the inference without re-doing the work. The convention throughout report TDR-2026-02 is that an absence of evidence is reported as such rather than as evidence of absence. Nothing in the report at TDR-2026-02 rests on a test the reviewer designed; the tests were the company's, and the reviewer read the results and the method. L. Casaubon did not visit the two most recently commissioned sites, so the commissioning quality there is taken from the company's records alone. The report at TDR-2026-02 is one input to a diligence process and is not a substitute for the company's own engineering judgement.

The present report carries one structural limitation: the company's own metrics are the only source for several objectives, and the reviewer could not build an independent measurement inside the engagement. The present report relies on the company's incident tracker for completeness of the incident population, which the reviewer could corroborate but not prove. Report TDR-2026-02 did not rely on management representation alone; every material claim was traced to a document, a measurement or an observation. L. Casaubon led the work behind this report, combining document review, 16 interviews and hands-on inspection of running systems. Wrenfield Technical Advisory LLP has costed remediation at the company's own loaded engineering rate rather than at a market rate.

The reviewer found the interview evidence for the present report candid, and in two cases more candid than the written record. 7 teams contributed to the interview programme for the present report, and one of them was interviewed twice after a document arrived late. The reviewer has resisted the temptation to tidy report TDR-2026-02: the open items are open because the evidence is not there yet. This report preserves 7 unresolved questions rather than resolving them by assumption. The capacity position, load-test evidence and latency behaviour of the platform at fleet scale is a wide field, and this report is explicit about which parts of it were reached.

Schedule J - Site deployments, capacity headroom and integration target (TDR-2026-02)

Site	Customer	Robots deployed	Go-live	Capacity headroom	Warehouse system	Note
Fremont, California	internal engineering site	34	2021-06	not measured	test harness	the reference site, and the only one with a laboratory alongside
Meridian, Idaho	Asterion Logistics	96	2022-09	22%	vendor A	the longest-running customer deployment and the source of the most incident history
Grand Prairie, Texas	Asterion Logistics	78	2023-04	31%	vendor A	the second Asterion site, commissioned on the same template
Wilmington, Ohio	Crestline Fulfillment	112	2023-08	14%	vendor B	the largest deployment and the closest to the allocator scaling limit
Lehigh Valley, Pennsylvania	Crestline Fulfillment	64	2024-02	38%	vendor B	commissioned during the connector rework
Ontario, California	Evergreen Warehousing	88	2024-05	26%	vendor C	the site whose shift pattern change caused the April 2025 charge incident
Groveport, Ohio	Granite Peak Distribution	52	2024-09	44%	vendor A	opportunity charging only, which is where the pack estimator drift appeared
Kenosha, Wisconsin	Dunbar Industrial	41	2024-11	49%	vendor D	the second opportunity-charging site
Tracy, California	Harborline Systems	58	2025-01	36%	vendor B	the site whose cleaning schedule now overlaps the peak shift
Shepherdsville, Kentucky	Fathom Mobility	47	2025-03	41%	vendor C	manipulator-equipped robots are concentrated here
Reno, Nevada	Blue Mesa Components	29	2025-05	57%	vendor D	the smallest customer deployment
Newnan, Georgia	Ironwood Grocery Group	66	2025-07	33%	vendor B	narrow-aisle geometry, and the source of the August 2025 mutual-yield stall
Whitestown, Indiana	Juniper Bay Apparel	44	2025-09	45%	vendor C	the site with the standing map publication workaround
Bethlehem, Pennsylvania	Kestrel Medical Supply	37	2025-11	51%	vendor A	sparse survey features, which is why relocalisation is slow here
Chino, California	Larkspur Home	33	2026-01	54%	vendor D	the go-live delayed by the forked provisioning template
Edwardsville, Illinois	Crestline Fulfillment	24	2026-03	63%	vendor B	the newest Crestline site, still ramping
Lockbourne, Ohio	Granite Peak Distribution	19	2026-04	68%	vendor A	commissioned after the template

Site	Customer	Robots deployed	Go-live	Capacity headroom	Warehouse system	Note
						baseline check landed
Mesquite, Texas	Evergreen Warehousing	16	2026-06	71%	vendor C	in acceptance testing at the review date

2. Load-test programme and the LT-2026-11 result

Scenario LT-2026-11, taken with the other runs in the series, shows that the pass verdict depends on a target that was last recalibrated in 2024. Comparing load run LT-2026-11 with its neighbour in the series, the difference is the order rate rather than the fleet size. Before reading scenario LT-2026-11, note that the environment carries the production cloud configuration and a laboratory network profile. run LT-2026-11 is the clearest instance in TDR-2026-02 of a control that exists on paper and is weaker in practice.

The percentile set for run LT-2026-12 reads 1.4 s to console, 22 s and 48 s, with errors at 0.08%. On 2026-04-09 the team ran scenario LT-2026-12, the telemetry ingest at twice the current fleet case, at 2400 robots. Telemetry ingest at twice the current fleet run LT-2026-12 shows where the system runs out of headroom: the ingest path scales horizontally with no shared bottleneck. telemetry ingest at twice the current fleet run LT-2026-12 is the clearest instance in TDR-2026-02 of engineering practice being better than the documentation suggests.

The environment behind scenario LT-2026-13 carries the current production topology with one region held out, and the reviewer weighted the result accordingly. LT-2026-13 is recorded as fail. The target scenario LT-2026-13 was measured against is the thirty-second telemetry visibility objective. 1440-robot run LT-2026-13 carries a follow-up request that was satisfied after the draft of TDR-2026-02 was circulated.

Four hour edge buffer drain after an outage run LT-2026-14 is best read against the neighbouring run in the same series, from which it differs in the size of the concurrent operator session pool. Weighed against the projected fleet, 720-robot run LT-2026-14 indicates that the improvement between the two runs is real and its durability is untested under a shard rebalance. 720 robots and 13,800 orders an hour define four hour edge buffer drain after an outage run LT-2026-14, run on 2026-04-15. 720-robot run LT-2026-14 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

The point of map publication to a thousand robots run LT-2026-15 within the series is the order rate rather than the fleet size. Weighed against the projected fleet, LT-2026-15 indicates that the result is consistent with the incident record, which is the strongest corroboration available. Scenario LT-2026-15 identified the binding constraint: inside the fifteen-minute objective at the tested fleet size. 1000-robot run LT-2026-15 was checked against the corresponding entry in the company's own register, and the register was current.

The result recorded for 720-robot run LT-2026-16 is pass with observation, which the reviewer traced to the underlying measurement. In 720-robot run LT-2026-16 the bottleneck was offline robots applied on reconnect, which is the behaviour the February 2026 incident exposed. On 2026-04-17 the team ran LT-2026-16, the config distribution with thirty per cent of robots offline case, at 720 robots. scenario LT-2026-16 is the reason one section of TDR-2026-02 concludes narrowly rather than generally.

The environment behind load run LT-2025-11 carries a fleet simulator rather than physical robots, with the control loop unchanged, and the reviewer weighted the result accordingly. Load run LT-2025-11 supports this conclusion: the pass verdict depends on a target that was last recalibrated in 2024. Against the stated targets, load run LT-2025-11 is pass. run LT-2025-11 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

The environment behind run LT-2025-12 carries a replayed order stream taken from the two largest sites, and the reviewer weighted the result accordingly. Load run LT-2025-12 was marked pass by the performance team. 120-robot run LT-2025-12 exercised the single site, peak hour case with 120 robots and 2,640 orders an hour, on 2025-10-09. 120-robot run LT-2025-12 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

What scenario LT-2025-13 does and does not establish is worth stating: the error rate rather than the latency is the term that would first breach a customer commitment. The percentile set for single site, peak with a charger outage run

LT-2025-13 reads 310 ms, 870 ms and 1,690 ms, with errors at 0.11%. Applying the thirty-second telemetry visibility objective to 120-robot run LT-2025-13 produces the verdict recorded above. LT-2025-13 is reported at the evidence strength available and has not been promoted to fit the narrative.

Schedule P - Load-test matrix against the published latency targets (TDR-2026-02)

Run	Scenario	Concurrent robots	Orders per hour	p50	p95	p99	Error rate	Verdict
LT-2025-11	single site, steady state	120	1,850	210 ms	480 ms	760 ms	0.02%	pass
LT-2025-12	single site, peak hour	120	2,640	265 ms	690 ms	1,240 ms	0.05%	pass
LT-2025-13	single site, peak with a charger outage	120	2,640	310 ms	870 ms	1,690 ms	0.11%	pass with observation
LT-2025-14	two sites, shared region	240	4,900	290 ms	820 ms	1,510 ms	0.09%	pass
LT-2025-15	four sites, shared region	480	9,400	355 ms	1,120 ms	2,240 ms	0.18%	pass with observation
LT-2025-16	six sites, shared region	720	13,800	470 ms	1,680 ms	3,410 ms	0.34%	fail against the 1,200 ms p95 target
LT-2025-17	order burst, ten times nominal for two minutes	240	12,000	620 ms	2,350 ms	5,900 ms	1.24%	fail against the 1,200 ms p95 target
LT-2025-18	telemetry burst with control traffic on the shared topic	480	9,400	540 ms	4,100 ms	11,300 ms	0.42%	fail
LT-2025-19	region failover during peak	480	9,400	410 ms	1,950 ms	6,700 ms	2.85%	fail against the error rate target
LT-2025-20	database failover, allocator primary	240	4,900	340 ms	1,410 ms	4,200 ms	0.96%	pass with observation
LT-2026-01	six sites after the topic split	720	13,800	395 ms	1,090 ms	2,050 ms	0.14%	pass
LT-2026-02	eight sites, shared region	960	18,200	455 ms	1,340 ms	2,880 ms	0.27%	fail against the 1,200 ms p95 target
LT-2026-03	eight sites with the allocator read replicas enabled	960	18,200	400 ms	1,150 ms	2,310 ms	0.21%	pass
LT-2026-04	ten sites, shared region	1200	22,600	520 ms	1,720 ms	4,050 ms	0.48%	fail against the 1,200 ms p95 target
LT-2026-05	order burst after the broker queue change	480	24,000	480 ms	1,240 ms	2,760 ms	0.31%	pass with observation
LT-2026-06	shard rebalance under peak load	720	13,800	430 ms	1,280 ms	3,120 ms	0.36%	pass with observation
LT-2026-07	identity service token storm	960	18,200	180 ms	520 ms	1,010 ms	0.03%	pass
LT-2026-08	console session concurrency	480	9,400	290 ms	740 ms	1,480 ms	0.07%	pass
LT-2026-09	console session concurrency at twice the plan	480	9,400	380 ms	1,460 ms	3,900 ms	0.52%	fail

Run	Scenario	Concurrent robots	Orders per hour	p50	p95	p99	Error rate	Verdict
LT-2026-10	warehouse connector burst, one tenant	240	8,800	610 ms	1,830 ms	4,400 ms	0.68%	fail against the 1,500 ms p95 acknowledgement target
LT-2026-11	warehouse connector burst after outbox partitioning	240	8,800	440 ms	1,210 ms	2,540 ms	0.29%	pass
LT-2026-12	telemetry ingest at twice the current fleet	2400	n/a	1.4 s to console	22 s	48 s	0.08%	pass
LT-2026-13	twelve sites, projected 2027 fleet	1440	27,000	690 ms	2,450 ms	6,100 ms	0.74%	fail
LT-2026-14	four hour edge buffer drain after an outage	720	13,800	not applicable	not applicable	not applicable	0.00%	pass with observation
LT-2026-15	map publication to a thousand robots	1000	n/a	3 m 20 s	8 m 40 s	13 m 10 s	0.10%	pass
LT-2026-16	config distribution with thirty per cent of robots offline	720	13,800	1 m 50 s	4 m 30 s	9 m 20 s	0.00%	pass with observation

3. Latency percentiles against the published targets

Comparing LT-2025-14 with its neighbour in the series, the difference is the warehouse system endpoint used. Measurement of 240-robot run LT-2025-14 is against the 1,200 ms p95 objective published for order assignment rather than against a customer commitment. Scenario LT-2025-14 is recorded as pass. load run LT-2025-14 is reported here in full because a summary of it in an earlier draft of TDR-2026-02 was misread as an assurance.

The fifteen-minute map publication objective is the target applied to load run LT-2025-15. The environment behind load run LT-2025-15 carries a smaller order history than production and a synthetic warehouse endpoint, and the reviewer weighted the result accordingly. Scenario LT-2025-15 was marked pass with observation by the performance team. LT-2025-15 was inspected in the repository rather than described from the ticket, which is why the account is specific.

The production cloud configuration and a laboratory network profile describes the environment six sites, shared region run LT-2025-16 ran in. The constraint surfaced by scenario LT-2025-16 is shadow store shard hotspotting on the two largest sites. 1,680 ms at p95 is the figure that matters in LT-2025-16, against 470 ms median and 3,410 ms at the tail. 720-robot run LT-2025-16 was checked against the corresponding entry in the company's own register, and the register was current.

Order broker journal write queue depth exceeded its alert threshold was the constraint 240-robot run LT-2025-17 exposed. Load run LT-2025-17 sits against the 1,200 ms p95 objective published for order assignment; whether that target is the right one is a separate question. The result recorded for run LT-2025-17 is fail against the 1,200 ms p95 target, which the reviewer traced to the underlying measurement. LT-2025-17 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

480 robots and 9,400 orders an hour define LT-2025-18, run on 2025-10-22. Configuration of the environment for load run LT-2025-18 included a smaller order history than production and a synthetic warehouse endpoint. Measurement of LT-2025-18 is against the thirty-second telemetry visibility objective rather than against a customer commitment. telemetry burst with control traffic on the shared topic run LT-2025-18 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Measurement of scenario LT-2025-19 is against the 1,500 ms p95 acknowledgement objective for the warehouse connector rather than against a customer commitment. A replayed order stream taken from the two largest sites describes

the environment LT-2025-19 ran in. Run LT-2025-19 is the region failover during peak run, sized at 480 robots and 9,400 orders an hour. region failover during peak run LT-2025-19 rests on a single source, and the reviewer has said so rather than implying corroboration.

Reading scenario LT-2025-20 for the binding constraint gives recovery inside 40 s but with an assignment gap. What database failover, allocator primary run LT-2025-20 does and does not establish is worth stating: the measurement is credible and the extrapolation the company draws from it is not. Database failover, allocator primary run LT-2025-20 returned 340 ms median, 1,410 ms at the 95th percentile and 4,200 ms at the 99th, erroring at 0.96%. load run LT-2025-20 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Measured latency in six sites after the topic split run LT-2026-01 was 395 ms at the median, 1,090 ms at p95 and 2,050 ms at p99. The six sites after the topic split test, recorded as six sites after the topic split run LT-2026-01, was executed on 2026-02-11. Reading scenario LT-2026-01 for the binding constraint gives the control topic split removed the telemetry coupling. run LT-2026-01 was checked against the corresponding entry in the company's own register, and the register was current.

Eight sites, shared region run LT-2026-02 is the eight sites, shared region run, sized at 960 robots and 18,200 orders an hour. The environment behind 960-robot run LT-2026-02 carries a synthetic robot fleet driven from recorded mission traces, and the reviewer weighted the result accordingly. The percentile set for scenario LT-2026-02 reads 455 ms, 1,340 ms and 2,880 ms, with errors at 0.27%. run LT-2026-02 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Schedule H - Change volume, change failure rate and restore time by month (TDR-2026-02)

Month	Changes to production	Failed changes	Change failure rate	Mean time to restore	Rollbacks
2025-01	48	3	6.2%	38 min	2
2025-02	58	4	6.8%	44 min	2
2025-03	45	1	2.2%	54 min	0
2025-04	51	1	1.9%	58 min	3
2025-05	69	4	5.7%	25 min	3
2025-06	43	0	0.0%	82 min	2
2025-07	50	4	8.0%	85 min	0
2025-08	73	2	2.7%	86 min	1
2025-09	46	1	2.1%	26 min	2
2025-10	71	4	5.6%	41 min	2
2025-11	44	1	2.2%	58 min	2
2025-12	66	4	6.0%	25 min	0
2026-01	78	5	6.4%	24 min	0
2026-02	43	1	2.3%	33 min	3
2026-03	63	0	0.0%	72 min	2
2026-04	39	1	2.5%	37 min	2
2026-05	46	3	6.5%	80 min	2
2026-06	58	3	5.1%	46 min	2
2026-07	46	3	6.5%	64 min	3
2026-08	71	6	8.4%	22 min	0

4. The services that bound throughput

Order Broker releases weekly. Order Broker owns its own data and exposes it only through a versioned interface, which the reviewer read. Hosting for Order Broker is cloud, containers on a managed Kubernetes cluster, and its Go 1.22 code writes through to PostgreSQL 14 with an event journal. Order Broker sits between the WMS connector and the site configuration service and the fleet allocator and the analytics warehouse loader, and that position is what makes its availability objective load-bearing. The backlog for Order Broker is 11 items deep and contains an acknowledged defect whereby order cancellation races with assignment and can leave an orphaned task record. Order Broker is reported at the evidence strength available and has not been promoted to fit the narrative.

What the reviewer had for Order Broker was the broker journal schema, the nightly reconciliation output and two cancellation incidents; the conclusions below do not reach past it. Service-level intent for Order Broker states that no accepted order is lost, measured by journal reconciliation each night. Ownership of Order Broker sits with Fleet Services; Grace Lindqvist signs off changes to it. Supportability of Order Broker turns on one fact: none identified. Automated test

coverage on Order Broker stands at 78 per cent line coverage. Order Broker would be a candidate for the first post-close engineering review under any reasonable plan.

The costmap fusion pipeline and the localization engine feed Motion Planner; a failure in Motion Planner is felt by the traffic manager and the on-robot drive supervisor. On paper Motion Planner releases every three weeks, promoted ring by ring, and the observed practice did not diverge from that. Motion Planner reports 71 per cent line coverage, measured by the same instrumentation the pipeline uses. For Motion Planner the reviewer worked from eleven design records, the planner benchmark suite and four weeks of production plan traces. Where Motion Planner is concerned, the commitment is that plan cycle completes inside 40 ms at the 99th percentile; the reviewer traced the metric to its source. Motion Planner is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Of the 14 open items against Motion Planner, the reviewer treats one as material: replan oscillation when two robots negotiate the same narrow-aisle pinch point. Interface discipline around Motion Planner is adequate; the schema is versioned and change is gated on a compatibility check. A local key-value snapshot flushed every ninety seconds is the system of record for Motion Planner, a Rust 1.79 component running on on-robot Linux with a real-time preemption patch. Placement of Motion Planner is edge, one instance per robot with no cloud fallback, which the reviewer tested by tracing a request end to end. Supportability of Motion Planner turns on one fact: none; the toolchain is pinned to a supported compiler release. Motion Planner is reported at the evidence strength available and has not been promoted to fit the narrative.

Underlying Speed and Separation Monitor in this report are the separation monitoring design note, the internal safety case and six bench approach tests. Safety Engineering holds Speed and Separation Monitor, and the ownership entry naming Bettina Vogel matches the on-call rotation. For Speed and Separation Monitor the reviewer records that the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. Speed and Separation Monitor is deployed edge, cross-checked against the safety controller's own inputs. For Speed and Separation Monitor the stated target is that commanded speed is reduced before the separation distance falls below the protective distance, expressed against production traffic rather than a benchmark. Speed and Separation Monitor appears twice in the working papers for TDR-2026-02 because the first answer was corrected after a follow-up request.

Implementation of Speed and Separation Monitor rests on C++17 with a safety-reviewed core; the process runs on on-robot Linux, pinned to the safety compute path and persists to a rolling in-memory window with fault snapshots to the safety controller. Speed and Separation Monitor carries 4 open defects, among them a case where the protective distance calculation uses a fixed operator approach speed rather than a measured one. 87 per cent line coverage is reported for Speed and Separation Monitor, and the reviewer re-ran the measurement on a clean checkout. The costmap fusion pipeline and the safety-rated scanner inputs feed Speed and Separation Monitor; a failure in Speed and Separation Monitor is felt by the safety controller and the motion planner speed limit. Interface discipline around Speed and Separation Monitor is adequate; the schema is versioned and change is gated on a compatibility check. Speed and Separation Monitor is the entry that changed most between the first and second document requests behind TDR-2026-02.

OTA Updater's dependency ageing was checked: none; artefact signing was moved to the current key format in 2025. OTA Updater does not share a schema with any consumer; every read by the on-robot supervisor and the fleet health monitor goes through the published interface. Deployment of OTA Updater is split edge and cloud, with staged rings and a per-site pause control, which sets the blast radius of any defect in it. Behind OTA Updater sits content-addressed object storage for artefacts, PostgreSQL 14 for state; the service itself is Rust 1.79 on the robot, Go 1.22 in the cloud on split: an on-robot agent and a cloud release service. The on-robot supervisor and the fleet health monitor rely on OTA Updater, which in turn relies on the release orchestrator and the artefact signing service. OTA Updater would move from a concern to a verified defect if one further measurement were supplied.

Ownership of OTA Updater sits with Platform Infrastructure; Erik Palmgren signs off changes to it. OTA Updater commits that an update is either fully applied or fully rolled back, with no mixed-version robot, and the objective is measured continuously rather than sampled. The cadence recorded for OTA Updater is every three weeks, following the autonomy release, which matches the observed release history. Automated test coverage on OTA Updater stands at 76 per cent line coverage. Documents and data inspected for OTA Updater were the update state machine specification, signing key custody notes and two rollback drills. OTA Updater would move from a concern to a verified defect if one further measurement were supplied.

Platform Infrastructure publishes one objective for Event Bus: publish acknowledgement inside 25 ms at p95 and no partition without a leader. What the reviewer had for Event Bus was the cluster configuration export, topic inventory and a partition failover exercise; the conclusions below do not reach past it. On Event Bus the conclusion is qualified: the objective is met in production, but the evidence that it will be met after the next dependency upgrade is weak. Interface discipline around Event Bus is adequate; the schema is versioned and change is gated on a compatibility check. Event Bus sits at no meaningful automated unit coverage, verification being by witnessed test instead; the trend across the reviewed period was down by two points after a large refactor. Event Bus is the reason one section of TDR-2026-02 concludes narrowly rather than generally.

Event Bus carries 4 open defects, among them a case where one topic carries both control and telemetry traffic, so a telemetry backlog delays control events. Event Bus is owned by the Platform Infrastructure team, with Erik Palmgren named as the engineering owner of record. The ageing position for Event Bus is the broker minor release in production leaves the vendor's support window in 2027, and the reviewer checked it against the vendor statements on file. Event Bus pairs a operated open-source broker, no in-house code implementation with cloud, a managed streaming cluster, and its data lives in the broker's own log storage with seven-day retention. On paper Event Bus releases upgraded twice a year, and the observed practice did not diverge from that. Event Bus is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Data ownership is clear for Remote Operations Console: it writes PostgreSQL 14 for session records and object storage for clips and no other service writes there. Against Remote Operations Console, the coverage figure is 52 per cent line coverage, which coverage alone overstates the assurance here, because the bench suite is the real safety net. Remote Operations Console runs cloud, with a thin session relay in each region; that placement is what determines whether a fault is one robot or one region. The Remote Operations Console codebase is TypeScript with a React front end throughout, deployed onto cloud, a static bundle plus a session gateway, and its durable record is PostgreSQL 14 for session records and object storage for clips. Responsibility for Remote Operations Console was confirmed with Anjali Sekhon, whose team, Applications, also holds the on-call duty. Remote Operations Console was checked against the company's own register, which had not been updated since the change landed.

27 items sit in the Remote Operations Console backlog; the entry that bears on the investment case is that the session list keeps rendering a robot that has already left the site until a manual refresh. The reviewer's judgement on Remote Operations Console is that the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. Reading the dependency map from Remote Operations Console outward gives the identity service, the telemetry ingest path and the video snippet service on the input side and the traffic manager and the alarm router on the output side. Release cadence for Remote Operations Console is twice a week, and the release records bear that out. The published objective for Remote Operations Console is that operator input reaches the robot inside 350 ms at p95 on a supported network. Remote Operations Console is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Underlying WMS Connector in this report are six tenant integration specifications and a month of connector error logs. The Integrations team carries WMS Connector on its service inventory, and Tomas Villareal answered for it in interview. Interface discipline around WMS Connector is adequate; the schema is versioned and change is gated on a compatibility check. WMS Connector is promoted every two weeks, per tenant on request, with the ring order fixed by policy. WMS Connector commits that order acknowledgement returned to the warehouse system inside 1.5 s at p95, and the objective is measured continuously rather than sampled. WMS Connector is the entry that changed most between the first and second document requests behind TDR-2026-02.

WMS Connector is written in Java 21, executes on cloud, containers on a managed Kubernetes cluster, and keeps its state in PostgreSQL 14 plus a message outbox table. WMS Connector carries 23 open defects, among them a case where duplicate order suppression relies on a customer-supplied identifier that two tenants reuse. Coverage for WMS Connector was 69 per cent line coverage at the review date, and the trend over the reviewed period was up by four points. The dependency map places the customer warehouse management systems and the identity service above WMS Connector and the order broker and the billing usage meter below it. WMS Connector's dependency ageing was checked: one customer integration still runs on a connector build from 2024. WMS Connector is included because a reader of TDR-2026-02 should be able to disagree with the inference without repeating the work.

On Telemetry Ingest the conclusion is qualified: the component is adequate today and carries a supportability liability that is not on the funded plan. Telemetry Ingest depends on the edge gateway agent and the event bus and is depended on by the fleet health monitor, the analytics warehouse loader and the alarm router. Automated test coverage on Telemetry Ingest stands at 61 per cent line coverage. For Telemetry Ingest the stated target is that telemetry visible in the operator console inside 30 s of emission at p95, expressed against production traffic rather than a benchmark. Underlying Telemetry Ingest in this report are the telemetry schema registry, retention configuration and a forced-offline buffer test. Telemetry Ingest was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Interface discipline around Telemetry Ingest is adequate; the schema is versioned and change is gated on a compatibility check. The manifests place Telemetry Ingest cloud, one ingest endpoint per region with an edge buffer on the robot. Farida Cisse of Data and Telemetry maintains Telemetry Ingest, and the service inventory entry was current at the review date. Telemetry Ingest ships weekly, with schema changes gated on a compatibility check; the reviewer sampled the last twelve releases. Of the 19 open items against Telemetry Ingest, the reviewer treats one as material: the edge buffer drops the oldest samples after four hours offline without raising an alert. Telemetry Ingest was one of two items where TDR-2026-02 could not obtain the underlying data within the engagement.

Schedule D - Automated coverage trend on the reviewed services (TDR-2026-02)

Service	Coverage 2025-09	Coverage 2026-03	Coverage at review	Change over period	Owning team
Motion Planner	63%	70%	71%	+8	Autonomy Core
Costmap Fusion	51%	55%	58%	+7	Perception
Fleet Allocator	74%	79%	82%	+8	Fleet Services
Traffic Manager	65%	72%	74%	+9	Fleet Services
Charge Scheduler	54%	60%	63%	+9	Fleet Services
Safety Controller	82%	88%	91%	+9	Safety Engineering
E-Stop Chain	n/a	n/a	witnessed test	not comparable	Safety Engineering
Lidar Driver	37%	44%	44%	+7	Robot Firmware
Localization Engine	61%	66%	66%	+5	Autonomy Core
WMS Connector	60%	68%	69%	+9	Integrations
Order Broker	72%	75%	78%	+6	Fleet Services
Telemetry Ingest	55%	58%	61%	+6	Data and Telemetry
OTA Updater	67%	73%	76%	+9	Platform Infrastructure
Identity and Authorization	77%	82%	84%	+7	Security Engineering
Remote Operations Console	48%	51%	52%	+4	Applications
Simulation Harness	44%	46%	47%	+3	Developer Productivity
Hardware-in-the-Loop Rig	35%	36%	39%	+4	Robot Firmware

4. The services that bound throughput - part 2 of 2

Where the 4. The services that bound throughput was interrupted by the page, it picks up again below, and the numbering is continuous throughout the schedule. Pagination alone determines where it falls.

The measurement basis recorded when the 4. The services that bound throughput opened is unchanged, so any figure here is comparable with any figure in the earlier parts. Pagination alone determines where it falls.

Schedule D - Automated coverage trend on the reviewed services (TDR-2026-02) (continued, part 2 of 2)

Service	Coverage 2025-09	Coverage 2026-03	Coverage at review	Change over period	Owning team
Fleet Health Monitor	50%	54%	57%	+7	Field Reliability
Battery Management Service	45%	51%	54%	+9	Robot Firmware
Gripper Controller	38%	38%	41%	+3	Robot Firmware
Map Authoring Studio	43%	48%	49%	+6	Applications
Site Provisioning Service	55%	62%	64%	+9	Platform Infrastructure
Event Bus	n/a	n/a	witnessed test	not comparable	Platform Infrastructure
Config Distribution	68%	69%	72%	+4	Platform Infrastructure

Service	Coverage 2025-09	Coverage 2026-03	Coverage at review	Change over period	Owning team
Robot Shadow Store	63%	68%	68%	+5	Fleet Services
Analytics Warehouse Loader	42%	43%	45%	+3	Data and Telemetry
Billing Usage Meter	75%	77%	79%	+4	Data and Telemetry
Alarm Router	46%	50%	51%	+5	Field Reliability
Video Snippet Service	37%	43%	43%	+6	Applications
Speed and Separation Monitor	82%	87%	87%	+5	Safety Engineering
Wheel Odometry Filter	53%	60%	62%	+9	Robot Firmware
Payload Scale Interface	29%	33%	36%	+7	Robot Firmware
Edge Gateway Agent	67%	73%	74%	+7	Platform Infrastructure
Release Orchestrator	47%	53%	55%	+8	Developer Productivity

5. Capacity headroom site by site

For Lockbourne, Ohio the material fact is that it is commissioned after the template baseline check landed. Lockbourne, Ohio runs 19 robots for Granite Peak Distribution and opened in 2026-04. Lockbourne, Ohio carries a follow-up request that was satisfied after the draft of TDR-2026-02 was circulated.

Against its peak profile, Whitestown, Indiana retains 45% headroom on a vendor C integration. The Whitestown, Indiana deployment serves Juniper Bay Apparel with 44 robots, live since 2025-09. Whitestown, Indiana was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

2025-05 is the go-live month at Reno, Nevada, where Blue Mesa Components now runs 29 robots. Reno, Nevada is the smallest customer deployment, and the reviewer weighted the sample accordingly. Reno, Nevada is the clearest instance in TDR-2026-02 of engineering practice being better than the documentation suggests.

Groveport, Ohio is notable in this review because it is opportunity charging only, which is where the pack estimator drift appeared. The Groveport, Ohio site integrates with vendor A and shows 44% headroom. Groveport, Ohio is reported at the evidence strength available and has not been promoted to fit the narrative.

Tracy, California is the site whose cleaning schedule now overlaps the peak shift, and the reviewer weighted the sample accordingly. 58 robots are deployed at Tracy, California, which went live in 2025-01. Tracy, California is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

On Newnan, Georgia the record shows that it is narrow-aisle geometry, and the source of the August 2025 mutual-yield stall. Ironwood Grocery Group operates 66 Helioscope robots at Newnan, Georgia, commissioned in 2025-07. Newnan, Georgia has no single accountable owner, and that is the finding the reviewer would press hardest.

Mesquite, Texas reports 71% headroom against its integration with vendor C. 2026-06 is the go-live month at Mesquite, Texas, where Evergreen Warehousing now runs 16 robots. Mesquite, Texas was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Crestline Fulfillment operates 112 Helioscope robots at Wilmington, Ohio, commissioned in 2023-08. Wilmington, Ohio is notable in this review because it is the largest deployment and the closest to the allocator scaling limit. Wilmington, Ohio was one of two items where TDR-2026-02 could not obtain the underlying data within the engagement.

Site ST-10 is Shepherdsville, Kentucky: Fathom Mobility, 47 robots, live from 2025-03. Headroom at Shepherdsville, Kentucky stands at 41%; vendor C is the integrated warehouse system. Shepherdsville, Kentucky has no single accountable owner, and that is the finding the reviewer would press hardest.

Ontario, California is the site whose shift pattern change caused the April 2025 charge incident, which is why the reviewer sampled it. 2024-05 is the go-live month at Ontario, California, where Evergreen Warehousing now runs 88 robots. Ontario, California was one of two items where TDR-2026-02 could not obtain the underlying data within the engagement.

96 robots are deployed at Meridian, Idaho, which went live in 2022-09. Meridian, Idaho is the longest-running customer deployment and the source of the most incident history, and the reviewer weighted the sample accordingly. Meridian, Idaho was one of two items where TDR-2026-02 could not obtain the underlying data within the engagement.

The Fremont, California site integrates with test harness and shows not measured headroom. Fremont, California is notable in this review because it is the reference site, and the only one with a laboratory alongside. Fremont, California was checked against the corresponding entry in the company's own register, and the register was current.

Site ST-03 is Grand Prairie, Texas: Asterion Logistics, 78 robots, live from 2023-04. Grand Prairie, Texas reports 31% headroom against its integration with vendor A. Grand Prairie, Texas illustrates why TDR-2026-02 separates the severity of a finding from the confidence behind it.

Bethlehem, Pennsylvania was included in the sample because it is sparse survey features, which is why relocalisation is slow here. Against its peak profile, Bethlehem, Pennsylvania retains 51% headroom on a vendor A integration. Bethlehem, Pennsylvania is the reason one section of TDR-2026-02 concludes narrowly rather than generally.

Edwardsville, Illinois came online in 2026-03 and now carries 24 robots for Crestline Fulfillment. The reason Edwardsville, Illinois appears here is that it is the newest Crestline site, still ramping. Edwardsville, Illinois is the clearest instance in TDR-2026-02 of engineering practice being better than the documentation suggests.

Lehigh Valley, Pennsylvania reports 38% headroom against its integration with vendor B. For Lehigh Valley, Pennsylvania the material fact is that it is commissioned during the connector rework. Lehigh Valley, Pennsylvania was one of two items where TDR-2026-02 could not obtain the underlying data within the engagement.

The Chino, California deployment serves Larkspur Home with 33 robots, live since 2026-01. For Chino, California the material fact is that it is the go-live delayed by the forked provisioning template. Chino, California is reported here in full because a summary of it in an earlier draft of TDR-2026-02 was misread as an assurance.

6. Capacity incidents and what they revealed

For INC-2025-124 the follow-through was tested rather than assumed, and landed on 2025-10-30; the load test was added on 2025-11-12. The affected population in multiple sites incident INC-2025-124 was 5 sites and 178 robots. The reviewer grades the INC-2025-124 evidence as moderate on the cause and strong on the delivery position, which was checked in the repository and has not upgraded the finding to compensate. What went wrong in incident INC-2025-124 was that the rebalance path bypassed the staleness annotation applied on the normal read path. The detection path for Robot Shadow Store incident INC-2025-124 was the console latency alert fired, though the underlying cause took a day to attribute. incident INC-2025-124 carries a follow-up request that was satisfied after the draft of TDR-2026-02 was circulated.

Robot Shadow Store incident INC-2025-124 would have been shorter or smaller had it not been that no test exercised a shard rebalance while the cluster was under load. Duration for multiple sites incident INC-2025-124 splits into 1 h 08 m of impact and 29 days to close the underlying cause. Robot Shadow Store incident INC-2025-124 was raised on 2025-10-01 after a shard rebalance served shadow state up to ninety seconds stale without marking it stale. Drawing the evidence in consistency event INC-2025-124 together, the team's analysis was honest about a gap it could have concealed. The commitment register shows INC-2025-124 against an undertaking to annotate staleness on every read path and add a rebalance-under-load test to the platform suite, with Marta Kellerman named and 2025-11-30 committed. multiple sites incident INC-2025-124 was raised again with the owning engineer after the initial interview, and the position did not change.

Process defect is the class of process defect incident INC-2025-146; Billing Usage Meter is the component. Billing Usage Meter incident INC-2025-146 was contained to no production site and no robot directly, the exposure being to internal systems. Process defect incident INC-2025-146 rests on moderate, resting on the team's own post-mortem and one supporting log extract evidence, which the reviewer states separately from the severity of the finding. INC-2025-146 took not applicable; caught before billing to stabilise and 11 days to close out. Incident INC-2025-146 was raised on 2025-11-19 after a re-run of a failed metering window double-counted usage for one customer. INC-2025-146 rests on a single source, and the reviewer has said so rather than implying corroboration.

In process defect event INC-2025-146, the runbook described the reset in prose rather than making it part of the command, which is a process finding rather than a code finding. The commitment register shows incident INC-2025-146 against an undertaking to fold the reset into the re-run command and make double-counting detectable in the ledger, with Nadia Semenova named and 2025-12-31 committed. Detection came through the monthly reconciliation caught it before invoicing, which places Larkspur Home incident INC-2025-146 on the reported side of the ledger. On delivery, incident INC-2025-146 shows this: landed on 2025-11-30. Incident INC-2025-146 was caused by a design gap: the re-run procedure required a manual ledger reset that the operator skipped. Larkspur Home incident INC-2025-146 appears twice in the working papers for TDR-2026-02 because the first answer was corrected after a follow-up request.

Incident INC-2026-015 rests on strong, resting on the production traces the team supplied evidence, which the reviewer states separately from the severity of the finding. How Config Distribution incident INC-2026-015 surfaced matters to the assessment: the on-robot supervisor raised a validation fault on the next mission. 2026-02-09 is the date of validation event INC-2026-015, on which a robot that had been offline at publication applied a configuration bundle without re-validating preconditions. Validation is the class of INC-2026-015; Config Distribution is the component. Contributing to validation incident INC-2026-015, and not repaired by the remediation, is that the offline apply path was added for a customer with intermittent connectivity and was lightly reviewed. incident INC-2026-015 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

The remediation committed after validation event INC-2026-015 was to move precondition validation to apply time and add an offline apply case to the platform suite, owned by Sam Oyelaran with a due date of 2026-03-31. Incident INC-2026-015 closure status is landed on 2026-03-03. Assessed on its own record, validation incident INC-2026-015 shows that the remediation addresses the immediate cause and leaves the contributing factor in place. Cause analysis in Fathom Mobility incident INC-2026-015 concluded that precondition validation ran at publication time rather than at apply time. INC-2026-015 touched 2 sites and 7 robots, and Fathom Mobility was the affected account. INC-2026-015 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

INC-2026-033 records that on 2026-03-25, a speed reduction engaged later than expected when an operator approached faster than the assumed walking speed. Verification of the INC-2026-033 remediation gives not landed; a design study is running and the vendor assessment is booked. Taken with the rest of the record, Asterion Logistics incident INC-2026-033 indicates that the customer-reported detection path is the finding, not the defect itself. The post-mortem for safety margin incident INC-2026-033 identifies the cause as follows: the protective distance uses a fixed assumed approach speed rather than a measured closing rate. The commitment register shows INC-2026-033 against an undertaking to measure closing rate where the sensing permits and re-run the protective distance calculation, with Bettina Vogel named and 2026-08-31 committed. safety margin event INC-2026-033 is reported at the evidence strength available and has not been promoted to fit the narrative.

Incident INC-2026-033 belongs to the safety margin group and sits with Speed and Separation Monitor. Time to mitigate INC-2026-033 was immediate speed reduction; 3 h to review the fleet, and time to resolve was open at the review date. Reported detection applies to INC-2026-033, since the on-robot safety log flagged a margin breach that did not become a contact event. Contributing to safety margin incident INC-2026-033, and not repaired by the remediation, is that the assumption was documented in the internal safety case and accepted at the time. INC-2026-033 touched one site and 6 robots, and Asterion Logistics was the affected account. Speed and Separation Monitor incident INC-2026-033 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Moderate on the cause and strong on the delivery position, which was checked in the repository is the evidence grade for Crestline Fulfillment incident INC-2026-039, and the reader should not read severity into it. The remediation committed after Crestline Fulfillment incident INC-2026-039 was to serialise cancellation against assignment and add an hourly orphan sweep, owned by Grace Lindqvist with a due date of 2026-05-31. Landed on 2026-04-26 is the position on the concurrency incident INC-2026-039 commitment as at the review date. Contributing to Crestline Fulfillment incident INC-2026-039, and not repaired by the remediation, is that the journal reconciliation runs nightly, so the orphan was invisible until the morning. Concurrency event INC-2026-039 traces back to a single fact: cancellation and assignment wrote to the task record without a shared lock. concurrency incident INC-2026-039 is reported at the evidence strength available and has not been promoted to fit the narrative.

Scope of impact in concurrency incident INC-2026-039 was one site and 1 robots, taken from the fleet records rather than from the narrative. Drawing the evidence in incident INC-2026-039 together, the elapsed resolution time reflects a scheduling decision rather than an engineering difficulty. Concurrency event INC-2026-039 reached the engineering team when the site supervisor reported a blocked location. Concurrency event INC-2026-039 belongs to the concurrency group and sits with Order Broker. On 2026-04-07 the record for Order Broker incident INC-2026-039 opens: an order cancellation raced with assignment and left an orphaned task that blocked a pick face for an hour. INC-2026-039 illustrates why TDR-2026-02 separates the severity of a finding from the confidence behind it.

Incident INC-2025-063 supports one conclusion and not another: the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. Fleet Allocator accounts for incident INC-2025-063, a policy defect event. Cause analysis in policy defect event INC-2025-063 concluded that the allocator filter excluded low-charge robots without checking whether a charge slot was actually available. For incident INC-2025-063 the follow-through was tested rather than assumed, and landed on 2025-06-08. The remediation committed after policy defect incident INC-2025-063 was to make the allocator consult charger availability before excluding a robot, owned by Marta Kellerman with a due date of 2025-07-31. Fleet Allocator incident INC-2025-063 is treated as material to the investment case rather than as a routine engineering item.

Reading Granite Peak Distribution incident INC-2025-063 from its first entry, on 2025-05-27 robots parked below the battery reserve threshold were never offered work and the site ran short of movers. Granite Peak Distribution felt policy defect event INC-2025-063, which affected 3 sites and 28 robots. Grading Granite Peak Distribution incident INC-2025-063 on evidence rather than on consequence gives moderate, resting on the team's own post-mortem and one supporting log extract. The two durations recorded for policy defect incident INC-2025-063 are 6 h 05 m to mitigate and 12 days to resolve. In policy defect event INC-2025-063, the charge scheduler and the allocator hold separate views of charger availability, which is a process finding rather than a code finding. policy defect event INC-2025-063 rests on a single source, and the reviewer has said so rather than implying corroboration.

The classification applied to Larkspur Home incident INC-2026-051 is configuration drift, and the affected component is Site Provisioning Service. In INC-2026-051, template drift was not detected because the baseline check compares only the default template, which is a process finding rather than a code finding. The judgement recorded against incident INC-2026-051 is that an operational workaround masked the defect for longer than the incident record suggests. The blast radius of INC-2026-051 was 1 site or tenant with no robot taken out of service. Cause analysis in incident INC-2026-051 concluded that the site template had been forked for a customer network requirement and drifted from the baseline. configuration drift incident INC-2026-051 would move from a concern to a verified defect if one further measurement were supplied.

Larkspur Home incident INC-2026-051 was raised on 2026-05-06 after a new site went live with a bespoke network template that omitted the outbound certificate pin. Reported detection applies to INC-2026-051, since the identity service rejected the site's first token request. Site Provisioning Service incident INC-2026-051 shows a short mitigation and a long tail: 2 h 18 m against 9 days. On Larkspur Home incident INC-2026-051 the evidence is limited, being a single account corroborated only by the ticket timeline; the severity assessment is stated separately below. Remediation for Larkspur Home incident INC-2026-051 is to check every template against the baseline and re-converge the four forked templates; Marcus Duplessis owns it and the committed date is 2026-06-15. configuration drift incident INC-2026-051 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

The trigger recorded for INC-2026-068 on 2026-06-16 was that odometry drift exceeded specification on a wet floor and localization confidence fell below the mission threshold. The reviewer's reading of environmental event INC-2026-068 is that management's severity rating is defensible and its remediation priority is not. The blast radius of Wheel Odometry Filter incident INC-2026-068 was one site and 14 robots. Contributing to environmental event INC-2026-068, and not repaired by the remediation, is that the site's cleaning schedule changed and now overlaps the peak shift. Not landed; the detector is prototyped and unvalidated is the position on the incident INC-2026-068 commitment as at the review date. INC-2026-068 illustrates why TDR-2026-02 separates the severity of a finding from the confidence behind it.

Grading environmental incident INC-2026-068 on evidence rather than on consequence gives limited; the log retention window had closed before the reviewer asked for the raw data. In environmental incident INC-2026-068 the first

signal was robots stopped and requested assistance, which the site reported as unplanned downtime, and the reviewer treats that as reported detection. What went wrong in INC-2026-068 was that the filter has no wet-floor slip model and does not flag the condition to the operator. Wheel Odometry Filter accounts for Harborline Systems incident INC-2026-068, a environmental event. Mitigation of incident INC-2026-068 took not a service outage; 3 h of degraded operation and full resolution took open at the review date. INC-2026-068 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Schedule F - Incident population examined, with follow-through status (TDR-2026-02)

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
INC-2025-041	2025-03-14	Traffic Manager	2	61	1 h 52 m	9 days	resource leak	landed
INC-2025-047	2025-04-02	Lidar Driver	1	12	38 m	3 days	timing	partly landed
INC-2025-052	2025-04-21	Charge Scheduler	1	44	4 h 10 m	16 days	scheduling	landed
INC-2025-058	2025-05-09	WMS Connector	2	0	2 h 35 m	21 days	data integrity	landed
INC-2025-063	2025-05-27	Fleet Allocator	3	28	6 h 05 m	12 days	policy defect	landed
INC-2025-071	2025-06-11	Telemetry Ingest	6	214	52 m	52 m	availability	landed
INC-2025-078	2025-06-24	Remote Operations Console	4	0	1 h 15 m	34 days	stale state	landed
INC-2025-084	2025-07-08	OTA Updater	1	11	27 m	5 days	state persistence	landed
INC-2025-090	2025-07-19	Safety Controller	1	3	immediate stop, 22 m to return to service	41 days	hardware fault	partly landed
INC-2025-096	2025-08-05	Alarm Router	0	0	not applicable; no service impact	26 days	process gap	partly landed
INC-2025-103	2025-08-22	Motion Planner	1	2	19 m	58 days	algorithm defect	partly landed
INC-2025-110	2025-09-03	Analytics Warehouse Loader	1	0	not applicable; reporting only	18 days	data integrity	landed
INC-2025-118	2025-09-17	Identity and Authorizatio n	0	0	3 h 40 m to revoke and rotate	7 days	secret exposure	partly landed
INC-2025-124	2025-10-01	Robot Shadow Store	5	178	1 h 08 m	29 days	consistency	landed
INC-2025-131	2025-10-14	Map Authoring Studio	1	34	3 h 22 m	44 days	process bypass	partly landed

6. Capacity incidents and what they revealed - part 2 of 2

This is part 2 of the 6. Capacity incidents and what they revealed, carried over unbroken; a reader reconciling a reference should treat the parts as a single register. The parts are to be read together.

Scope, cut-off and units for the 6. Capacity incidents and what they revealed were fixed at its first part and are not repeated, being unaltered. The sequence is unbroken either side of it.

Schedule F - Incident population examined, with follow-through status (TDR-2026-02) (continued, part 2 of 2)

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
INC-2025-139	2025-11-02	Event Bus	7	246	2 h 47 m	63 days	capacity	partly landed
INC-2025-146	2025-11-19	Billing Usage Meter	0	0	not applicable; caught	11 days	process defect	landed

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
					before billing			
INC-2025-152	2025-12-04	Video Snippet Service	0	0	not applicable; retention only	37 days	data retention	partly landed
INC-2026-004	2026-01-13	Localization Engine	1	18	not a service outage; 5 days to a workaround	open at the review date	performance	not landed
INC-2026-009	2026-01-27	Gripper Controller	1	4	1 h 30 m to quarantine the affected robots	48 days	calibration	landed
INC-2026-015	2026-02-09	Config Distribution	2	7	44 m	22 days	validation	landed
INC-2026-021	2026-02-24	Simulation Harness	0	0	not applicable; developer tooling	open at the review date	tooling	not landed
INC-2026-028	2026-03-10	Fleet Health Monitor	1	22	not a service outage; 2 days of delayed repair	31 days	observability	landed
INC-2026-033	2026-03-25	Speed and Separation Monitor	1	6	immediate speed reduction; 3 h to review the fleet	open at the review date	safety margin	not landed
INC-2026-039	2026-04-07	Order Broker	1	1	1 h 04 m	19 days	concurrency	landed
INC-2026-045	2026-04-21	Edge Gateway Agent	1	39	5 h 12 m	26 days	observability	partly landed
INC-2026-051	2026-05-06	Site Provisioning Service	1	0	2 h 18 m	9 days	configuration drift	partly landed
INC-2026-057	2026-05-19	Battery Management Service	2	5	not a service outage; robots derated	open at the review date	modelling	not landed
INC-2026-062	2026-06-02	Release Orchestrator	0	0	not applicable; process only	open at the review date	process control	not landed
INC-2026-068	2026-06-16	Wheel Odometry Filter	1	14	not a service outage; 3 h of degraded operation	open at the review date	environmental	not landed

7. The scalability programme, its sequence and its cost

Localization Engine item TD-13 is in design with a target date of 2026-12-31. In the programme register item TD-13 is behind a customer maintenance window that is not yet agreed. About TD-13 it should be said that carries the same dependency as the copyleft removal. TD-13 on Localization Engine is reported here in full because a summary of it in an earlier draft of TDR-2026-02 was misread as an assurance.

If quality item TD-06 is not done, the exposure is that it a nightly regression can reach a release candidate. About quality item TD-06 it should be said that blocked on the physics core determinism work. Quality item TD-06 is blocked with a target date of 2026-11-30. programme item TD-06 is reported here in full because a summary of it in an earlier draft of TDR-2026-02 was misread as an assurance.

Behind a platform change that has its own target date is where safety evidence item TD-18 sits in the programme. 0.75 engineer-quarters is the estimate on safety evidence item TD-18, which converts to $104,000. The plan treats register item TD-18 as funded only if the team's headcount plan is met, which the reviewer checked against the headcount model. register item TD-18 is reported at the evidence strength available and has not been promoted to fit the narrative.

No advisory affects the current version, which is why it ranks low is the point the reviewer would put to management on security supportability item TD-08. Replace the archived charting library in the operator console is entered at programme item TD-08 as an security supportability item. Register item TD-08 matters because it no upstream security updates for a customer-facing component. TD-08 on Remote Operations Console rests on a single source, and the reviewer has said so rather than implying corroboration.

In progress is the status of register item TD-11, targeted at 2026-09-30. Site Provisioning Service item TD-11 carries this observation: two templates remain forked as at the review date. The plan treats configuration item TD-11 as not funded, and carried in the register regardless, which the reviewer checked against the headcount model. programme item TD-11 was one of two items where TDR-2026-02 could not obtain the underlying data within the engagement.

Exposure carried by TD-22 is that it calibration is lost on a compute module swap. TD-22 carries this observation: a small item that keeps returning as a field support cost. The estimate for register item TD-22 is 0.25 engineer-quarters and $42,000. register item TD-22 is the clearest instance in TDR-2026-02 of engineering practice being better than the documentation suggests.

Sequencing of programme item TD-24 matters: it is behind a customer maintenance window that is not yet agreed. The estimate for Costmap Fusion item TD-24 is 0.25 engineer-quarters and $28,000. TD-24 is committed with a target date of 2026-09-30. register item TD-24 illustrates why TDR-2026-02 separates the severity of a finding from the confidence behind it.

Delivery item TD-14 is sized at 0.5 engineer-quarters, or $88,000 at the company's loaded rate. The dependency structure places register item TD-14 in parallel with the release-process changes. The reviewer's comment on Hardware-in-the-Loop Rig item TD-14 is that requires capital for a third rig rather than engineering effort. register item TD-14 was raised again with the owning engineer after the initial interview, and the position did not change.

Programme item TD-01 is last in a three-item chain, so its date is the least reliable on the register, which is why the target date is later than the effort estimate implies. Effort on Fleet Allocator item TD-01 is estimated at 3.5 engineer-quarters, costed at $420,000. Whether register item TD-01 happens depends on funding, and it is funded from the second quarter onward. programme item TD-01 was raised again with the owning engineer after the initial interview, and the position did not change.

Item TD-17 on OTA Updater in the debt register is to enforce a two-person rule on OTA artefact signing, and it bears on OTA Updater. 2026-11-30 is the target for TD-17 on OTA Updater, whose status is in design. Funding for TD-17 on OTA Updater is unfunded and dependent on capital rather than engineering effort, and the register does not say so. TD-17 on OTA Updater is the clearest instance in TDR-2026-02 of a control that exists on paper and is weaker in practice.

The business consequence recorded against support item TD-03 is that it no upstream fixes for a component on the safety-adjacent path. The dependency structure places TD-03 behind a platform change that has its own target date. Whether Costmap Fusion item TD-03 happens depends on funding, and it is partly funded, with the second phase unbudgeted. TD-03 is the clearest instance in TDR-2026-02 of a control that exists on paper and is weaker in practice.

TD-07 is the item to pin the full simulation build configuration and enable a determinism check, classified under tooling. Funding for programme item TD-07 is funded in the current financial year, and the register does not say so. The business consequence recorded against TD-07 on Simulation Harness is that it non-reproducible test results undermine every simulation claim. register item TD-07 was checked against the corresponding entry in the company's own register, and the register was current.

The reviewer's qualification on programme item TD-12 is that the site alert landed first; the buffer extension needs a storage change on the compute module. Unfunded and dependent on capital rather than engineering effort is the funding position on Edge Gateway Agent item TD-12. Edge Gateway Agent item TD-12 carries an estimate of 1 engineer-quarter; the costed figure is $144,000. resilience item TD-12 was raised again with the owning engineer after the initial interview, and the position did not change.

The dependency structure places security item TD-16 after the determinism work in the simulation harness. TD-16 appears in the register and is partly funded, with the second phase unbudgeted. The consequence of leaving TD-16 on

Localization Engine undone is that it a stolen compute module discloses customer warehouse layouts. register item TD-16 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Progress on programme item TD-02 is recorded as committed against a 2026-12-31 date. 1.5 engineer-quarters is the estimate on Localization Engine item TD-02, which converts to $185,000. The business consequence recorded against programme item TD-02 is that it strong-copyleft exposure in shipped robot software plus an unsupported dependency. Localization Engine item TD-02 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Register item TD-15 matters because it one credential compromise affects paging everywhere. Reading the programme plan, programme item TD-15 is last in a three-item chain, so its date is the least reliable on the register. The security entry at security item TD-15 is to add per-region credentials for the paging provider. TD-15 is the entry that changed most between the first and second document requests behind TDR-2026-02.

TD-09 is on the register because it an unsupported broker underneath every service. Register item TD-09 is funded and already staffed in the engineering plan the reviewer was shown. Reading register item TD-09 against the funded plan, the upgrade requires a maintenance window at every site. programme item TD-09 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Programme item TD-05 is sequenced behind a platform change that has its own target date. The register shows TD-05 on Hardware-in-the-Loop Rig as committed, due 2026-09-30. TD-05 carries this observation: outstanding since the April 2025 lidar timing incident. register item TD-05 illustrates why TDR-2026-02 separates the severity of a finding from the confidence behind it.

8. Resilience runs: failover, rebalance and buffer drain

On 2026-02-13 the team ran scenario LT-2026-03, the eight sites with the allocator read replicas enabled case, at 960 robots. Outcome of LT-2026-03 was pass, and the reviewer read the raw result rather than the summary. Latency in scenario LT-2026-03 was 400 ms at p50, 1,150 ms at p95 and 2,310 ms at p99, with an error rate of 0.21%. run LT-2026-03 was checked against the corresponding entry in the company's own register, and the register was current.

Scenario LT-2026-04 shows where the system runs out of headroom: the allocator write path needs partitioning to go further. LT-2026-04 sits against the 1,200 ms p95 objective published for order assignment; whether that target is the right one is a separate question. Scenario LT-2026-04 is not read as a headline number: the test environment carried a smaller dataset than production, so the database term is understated here. ten sites, shared region run LT-2026-04 is the clearest instance in TDR-2026-02 of engineering practice being better than the documentation suggests.

480-robot run LT-2026-05 supports this conclusion: one run does not establish a trend, and this scenario was executed once. Order burst after the broker queue change run LT-2026-05 exercised the order burst after the broker queue change case with 480 robots and 24,000 orders an hour, on 2026-02-18. Set beside the other runs, order burst after the broker queue change run LT-2026-05 varies only in whether control and telemetry shared a topic. order burst after the broker queue change run LT-2026-05 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

The point of shard rebalance under peak load run LT-2026-06 within the series is the presence of a failure injection. Errors in shard rebalance under peak load run LT-2026-06 were 0.36%, with latency of 430 ms, 1,280 ms and 3,120 ms across the three percentiles. What run LT-2026-06 does and does not establish is worth stating: the pass verdict depends on a target that was last recalibrated in 2024. scenario LT-2026-06 was inspected in the repository rather than described from the ticket, which is why the account is specific.

Reading LT-2026-07 for the binding constraint gives token issuance was not a limiting term at any tested scale. Run LT-2026-07 ran against an environment with a replayed order stream taken from the two largest sites. LT-2026-07 is the identity service token storm run, sized at 960 robots and 18,200 orders an hour. run LT-2026-07 is one of the entries where TDR-2026-02 relies on the company's own measurement rather than an independent one.

Against the stated targets, console session concurrency run LT-2026-08 is pass. Measurement of load run LT-2026-08 is against the fifteen-minute map publication objective rather than against a customer commitment. The test bed for load run LT-2026-08 carries a fleet simulator rather than physical robots, with the control loop unchanged, which is the

main reason the reviewer does not treat it as production-equivalent. scenario LT-2026-08 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Against the stated targets, LT-2026-09 is fail. Scenario LT-2026-09 exists in the series to isolate the size of the concurrent operator session pool. The console session concurrency at twice the plan test, recorded as 480-robot run LT-2026-09, was executed on 2026-02-25. LT-2026-09 is included because a reader of TDR-2026-02 should be able to disagree with the inference without repeating the work.

Weighed against the projected fleet, load run LT-2026-10 indicates that the test environment carried a smaller dataset than production, so the database term is understated here. The fifteen-minute map publication objective is the target applied to run LT-2026-10. Production-sized data but a single availability zone describes the environment load run LT-2026-10 ran in. load run LT-2026-10 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

9. Dependencies with a capacity implication

The open question on geom2d-fusion is that dynamic linking preserves the weak-copyleft boundary; counsel confirmed the analysis in 2025. In the approval log, geom2d-fusion is approved 2023-08-04. The terms applying to geom2d-fusion are LGPL-2.1; the copyleft classification is weak. geom2d-fusion is the only entry in its group where the reviewer and the engineering team reached different conclusions.

An upstream project that is active but changes interface frequently attaches to physics-core-fork independently of its licence terms. Physics-core-fork reaches the customer only through Simulation Harness, and the reviewer confirmed there is no second path. Physics-core-fork carries this qualification: the fork diverges by nineteen patches and no upstream merge path is maintained. physics-core-fork is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The open question on http-router is that attribution present. OSS-022 records http-router 1.9.4 against Order Broker, cloud service routing. The distribution question for http-router turns on the fact that it is used cloud service routing. http-router was compared with the equivalent position at two other components, and it is neither the best nor the worst.

OSS-005 records tsdb-client 3.1.4 against Telemetry Ingest, telemetry writes from the ingest service. In the approval log, tsdb-client is approved 2024-09-11. Tracing tsdb-client to its source gives public registry, signature verified. tsdb-client was excluded from an earlier draft of TDR-2026-02 and reinstated after the reviewer obtained the supporting record.

In addition to its terms, warehouse-sim-assets brings no identified risk beyond ordinary version currency. Warehouse-sim-assets affects Simulation Harness; no other component reviewed depends on it. Warehouse-sim-assets carries the approval entry approved with conditions 2024-11-06; the reviewer looked for a corresponding decision record. warehouse-sim-assets would move from a concern to a verified defect if one further measurement were supplied.

The reviewer's note on nav-graph-solver is that the notice file is complete and reproduced in the robot documentation. The supply route for nav-graph-solver is public registry, signature verified. Nav-graph-solver appears at OSS-001 in version 2.4.1; the usage recorded is linked into the planner as a static library. nav-graph-solver is included because a reader of TDR-2026-02 should be able to disagree with the inference without repeating the work.

The supply route for pgbouncer-managed is public registry, signature verified. Pgbouncer-managed carries a practical obligation: to share alike any derived asset it distributes. Policy compliance for pgbouncer-managed rests on its approval entry, which is approved 2024-03-05. pgbouncer-managed is one of the entries where TDR-2026-02 relies on the company's own measurement rather than an independent one.

Approval for media-transcode reads approved with conditions 2024-08-16, and the reviewer checked the approving record. Media-transcode carries this qualification: the build must exclude the GPL-configured components and the reviewer verified the current build flags. Media-transcode is licensed under LGPL-2.1 with an optional GPL build, with copyleft exposure recorded as weak. media-transcode is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Beyond licensing, grid-render carries a single-maintainer dependency with no succession plan. Public registry; the commercial licence was purchased in 2024 is the provenance recorded for grid-render. In the approval log, grid-render is approved with conditions 2023-07-19. grid-render was excluded from an earlier draft of TDR-2026-02 and reinstated after the reviewer obtained the supporting record.

What the company must actually do about lidar-vendor-sdk is to publish the modified files it distributes. Against lidar-vendor-sdk the reviewer would add that two major versions behind current; the vendor's support statement covers the deployed version to 2027. In the approval log, lidar-vendor-sdk is approved 2023-05-22. lidar-vendor-sdk is included because a reader of TDR-2026-02 should be able to disagree with the inference without repeating the work.

Meeting the terms on yaml-parse means nothing beyond retaining the licence text with the source. MIT governs yaml-parse, which the inventory marks as no copyleft. Approved 2023-09-29 is the approval entry for yaml-parse. yaml-parse was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Usage of k8s-operator-framework 1.31.0 is cloud deployment controllers, and the inventory reference is OSS-028. Beyond licensing, k8s-operator-framework carries a fork that must be maintained locally for as long as it is used. Licence text and the actual usage of k8s-operator-framework were read together, and attribution present. k8s-operator-framework was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

On provenance, otel-instrument shows public registry, signature verified. Under Apache-2.0, otel-instrument is treated as no copyleft in the inventory. Otel-instrument sits inside Fleet Health Monitor and is therefore in the shipped product rather than in tooling. otel-instrument carries a follow-up request that was satisfied after the draft of TDR-2026-02 was circulated.

At OSS-017 the component is wms-adapter-sdk, pinned at 2.2.0 and used one customer's warehouse system adapter. Meeting the terms on wms-adapter-sdk means to exclude a particular build configuration from the shipped artefact. On provenance, wms-adapter-sdk shows supplied by the warehouse system vendor under a written licence. wms-adapter-sdk is the clearest instance in TDR-2026-02 of a control that exists on paper and is weaker in practice.

10. Security findings that constrain the capacity plan

Map Authoring Studio finding SEC-2025-04 is remediated, with a closure date of 2025-11-27. Moderate evidence supports assessment finding SEC-2025-04, namely that reproduced, and independently observed in a field incident. Finding SEC-2025-04 arrives as a Medium finding with a 5.4 score, and the reviewer did not adjust either. Residual risk after remediation of SEC-2025-04 is medium; the shortened publication path is still outstanding. finding SEC-2025-04 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

The finding at SEC-2025-03 against Remote Operations Console, dated 2025-09-18, is that remote operations session relay accepted an unpinned certificate chain. SEC-2025-03 against Remote Operations Console is assessed as follows: the finding was closed within the committed window and the closure evidence was inspected. Finding SEC-2025-03 is rated Medium with a numeric band of 6.2 on the tester's scoring scale. SEC-2025-03 against Remote Operations Console leaves residual risk that the reviewer assesses as low; pinning is applied at the gateway and the agent. assessment finding SEC-2025-03 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Finding SEC-2026-13 translates into the following exposure: a large payload degrades broker throughput. Substantiating Low-severity finding SEC-2026-13 is the fact that throughput degradation measured by the tester, graded limited, resting on a configuration read rather than on a demonstrated exploitation. Remediation status on assessment finding SEC-2026-13 is remediated; the record shows 2026-05-15. Low is the rating assessment finding SEC-2026-13 carries, at 3.3 on the assessor's own scale. assessment finding SEC-2026-13 was excluded from an earlier draft of TDR-2026-02 and reinstated after the reviewer obtained the supporting record.

The reviewer declines to soften SEC-2026-11: management's acceptance is defensible on the evidence and should be revisited before the fleet doubles. Gripper controller parameter store writable by any on-robot process is the substance of SEC-2026-11 against Gripper Controller, raised on 2026-04-19. Finding SEC-2026-11 is not fully retired by its remediation: medium; the safety controller limit is the compensating control. Gripper Controller finding SEC-2026-11

matters because a non-privileged process could alter a force limit parameter. assessment finding SEC-2026-11 illustrates why TDR-2026-02 separates the severity of a finding from the confidence behind it.

The residual position on Identity and Authorization finding SEC-2026-08 is low; role grants now require a second approver and are alerted, stated after mitigation rather than assuming full success. The practical consequence in SEC-2026-08 is that one administrator can grant themselves tenant-wide access. The finding at assessment finding SEC-2026-08, dated 2026-04-18, is that identity service admin role assignable without a second approval. Tracking SEC-2026-08 to closure gives a status of remediated and a date of 2026-06-30. assessment finding SEC-2026-08 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Open is the status of Medium-severity finding SEC-2026-07, dated committed 2026-10-31. Finding SEC-2026-07 translates into the following exposure: an unpatched laboratory host sits on the engineering network. Evidence behind Hardware-in-the-Loop Rig finding SEC-2026-07 is that operating system version observed on the rig, which the reviewer treats as strong. Residual exposure on assessment finding SEC-2026-07 reads medium; network segmentation is the only current control. Hardware-in-the-Loop Rig finding SEC-2026-07 was tested rather than accepted, and the test is described in the working papers behind TDR-2026-02.

Residual risk after remediation of Medium-severity finding SEC-2025-07 is medium; two forked templates were not re-issued until May 2026. SEC-2025-07 matters because the same secret was reusable at any site provisioned from that template. Remediated is the status of Medium-severity finding SEC-2025-07, dated 2026-01-16. The evidence for Medium-severity finding SEC-2025-07 is strong: the secret was recovered from a template export. assessment finding SEC-2025-07 has no single accountable owner, and that is the finding the reviewer would press hardest.

The reason SEC-2025-09 was not downgraded is that a single key holder could sign an unreviewed firmware artefact. SEC-2025-09 against OTA Updater produces this judgement: management's acceptance is defensible on the evidence and should be revisited before the fleet doubles. SEC-2025-09 against OTA Updater was graded Medium, band 6.3, by Redwater Security engagement RW-2025-3. Residual exposure on SEC-2025-09 reads medium; a two-person rule is designed but not enforced in the tool. Medium-severity finding SEC-2025-09 is one of the entries where TDR-2026-02 relies on the company's own measurement rather than an independent one.

Finding SEC-2026-18 arrives as a Low finding with a 3.4 score, and the reviewer did not adjust either. Impact of SEC-2026-18 against Battery Management Service is not theoretical; a diagnostic flood can slow pack telemetry reporting. Tracking assessment finding SEC-2026-18 to closure gives a status of open and a date of committed 2026-12-15. Assessment finding SEC-2026-18 is assessed as follows: the item is open past its committed date and no revised date has been agreed. finding SEC-2026-18 was one of two items where TDR-2026-02 could not obtain the underlying data within the engagement.

Read alongside the remediation record, finding SEC-2026-16 shows that the item is open past its committed date and no revised date has been agreed. Rating for SEC-2026-16 is Medium, band 5.8; the reviewer re-read the scoring rationale and accepted it. The remediation ledger shows SEC-2026-16 against Costmap Fusion as remediated, 2026-06-26. Medium-severity finding SEC-2026-16 records that costmap fusion diagnostic socket bound to all interfaces, discovered on 2026-04-22. finding SEC-2026-16 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

The reviewer declines to soften SEC-2026-14: the severity is right and the ownership is unclear, since two teams each believe the other holds it. Low-severity finding SEC-2026-14 was opened on 2026-04-21 and states that traffic manager admin action lacked an audit record. Moderate, because the demonstration was performed in a laboratory tenant rather than in production evidence supports assessment finding SEC-2026-14, namely that action performed and the audit trail checked. Remediated is the status of SEC-2026-14 against Traffic Manager, dated 2026-06-18. assessment finding SEC-2026-14 was cross-checked against the engineering record supplied for TDR-2026-02, and the two agree.

What assessment finding SEC-2026-15 enables is this: operator confusion rather than a direct security impact. SEC-2026-15 against Site Provisioning Service reduces to a residual risk of low; management judged the operational cost of enforcement higher than the risk. As at the review date SEC-2026-15 against Site Provisioning Service was accepted and the closure entry reads not planned. For SEC-2026-15 against Site Provisioning Service the reviewer notes that the

severity is right and the ownership is unclear, since two teams each believe the other holds it. assessment finding SEC-2026-15 was checked against the company's own register, which had not been updated since the change landed.

Assessment finding SEC-2025-05 is remediated, with a closure date of 2026-02-06. Substantiating assessment finding SEC-2025-05 is the fact that the tester submitted synthetic telemetry from an unenrolled host, graded moderate. Finding SEC-2025-05 was opened on 2025-09-19 and states that telemetry ingest accepted a robot identity without checking device attestation. Severity on SEC-2025-05 against Telemetry Ingest is High, scored 7.8 by the assessor. SEC-2025-05 against Telemetry Ingest is the clearest instance in TDR-2026-02 of a control that exists on paper and is weaker in practice.

11. Capacity qualifications and unresolved questions

Where this report cannot conclude, it says what evidence would let it conclude. 7 items in this report are open, two of which the reviewer would treat as pre-closing information requests rather than post-close work. The interview programme behind report TDR-2026-02 covered 7 teams and produced 17 follow-up requests, of which most were satisfied. L. Casaubon interviewed 16 engineers and managers for the present report; two declined to be quoted on one topic and that is noted where it matters. Wrenfield Technical Advisory LLP has not been asked to opine on valuation, and nothing in report TDR-2026-02 should be read as doing so.

Nothing in TDR-2026-02 is presented as verified on the strength of an interview alone. L. Casaubon has left several concerns in report TDR-2026-02 at limited evidence strength rather than dropping them or dressing them up. For Ardent Peak's purposes, the report at TDR-2026-02 concludes that the technical debt register is honest about what exists and optimistic about when it will be addressed. L. Casaubon would summarise the report at TDR-2026-02 in one sentence: the technical debt register is honest about what exists and optimistic about when it will be addressed. Wrenfield Technical Advisory LLP has flagged 3 matters for the transaction record and left the remainder as engineering work.

This report treats a backlog label as evidence of intent and not as evidence of remediation, which is why several items remain open in this report that the company regards as handled. The reviewer asked in each case whether a declined item had a compensating control, and this report reports the answer even where it is unsatisfactory. The report at TDR-2026-02 sampled 9 services from an inventory of 34, weighted towards those on the critical control path. For the report at TDR-2026-02 the reviewer sampled 6 sites, including the largest deployment and the newest. Any figure in this report that the reviewer could not trace to a system is attributed to the person who gave it.

The present report proposes 6 metrics for the post-close monitoring pack, each already produced by an existing system. Monitoring recommended by the present report relies on metrics the company already emits, so no new instrumentation is required to start. The open items in report TDR-2026-02 are grouped by what would resolve them rather than by which team owns them, because several cross team boundaries. L. Casaubon has dated each open item in the present report so that a reader can see which have been open longest rather than only how many there are. Readers comparing report TDR-2026-02 with an earlier technical report should note that the service inventory was renamed in between.

12. Conclusions on capacity against the operating plan

On the evidence available by 2026-02-18, TDR-2026-02 concludes that the technical debt register is honest about what exists and optimistic about when it will be addressed. Stated without qualification for once, the conclusion of the report at TDR-2026-02 is that the technical debt register is honest about what exists and optimistic about when it will be addressed. 2026-02-18 is the evidence cut-off for TDR-2026-02, whose subject is the capacity position, load-test evidence and latency behaviour of the platform at fleet scale. The instruction behind report TDR-2026-02 was narrow: examine the capacity position, load-test evidence and latency behaviour of the platform at fleet scale and report by 2026-02-18. This report does not restate the company's own reporting where the reviewer was able to verify the underlying data instead.

Wrenfield Technical Advisory LLP notes that 3 of the matters in this report are capable of being closed before completion if the company is asked in time. L. Casaubon would put 4 items from this report into the first hundred days rather than into a warranty. One limitation of the report at TDR-2026-02 should be stated plainly: the reviewer had no production access and relied on the company's own measurements for live latency. The present report does not include a

code audit; the reviewer read code where a specific claim required it and no further. The present report was prepared on the footing that a reader may be an engineer or may not, and defines terms where the distinction matters.

6 indicators are recommended in this report; the reviewer deliberately avoided proposing anything that needs a new pipeline. The report at TDR-2026-02 recommends that the remediation commitment register be reported to the board quarterly with evidence attached, not just status. The present report concludes that the technical debt register is honest about what exists and optimistic about when it will be addressed. L. Casaubon's conclusion in report TDR-2026-02 is that the technical debt register is honest about what exists and optimistic about when it will be addressed. The present report is one input to a diligence process and is not a substitute for the company's own engineering judgement.